exerc.py × Terminal

```
  # Generate and save the images
  def chain(prompts):
    for i, prompt in enumerate(prompts):
      response = openai.Image.create(
        prompt=prompt,
        n=1,
        size="256x256"
      )

      # Get the image URL from the response
      image_url = response['data'][0]['url']

      # Download and save the image
      img_data = requests.get(image_url).content
      with open(f"test{i+1}.jpg", 'wb') as handler:
          handler.write(img_data)
```

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Guide ×

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Coding Exercise

Write a function `chain()` that takes a list of prompts and generates images based on the prompts. The function should use chaining to generate the images. The function should save the images to the current directory, using the following convention: `test{i+1}.jpg`, where `i` is the index of the prompt in the list. Please make sure the images are `256x256`.

For example, if the list of prompts is `["A cat", "A dog", "A bird"]`, the function should generate three images:

- `test1.jpg`: An image of a cat
- `test2.jpg`: An image of a dog
- `test3.jpg`: An image of a bird
  The function should be able to handle any number of prompts in the list

TRY IT!

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Click here to refresh your image 3

When ready to submit your code, please click the button below.

```python
import os
import openai
import requests
import secret

openai.api_key=secret.api_key



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  for i, prompt in enumerate(prompts):
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    # Get the image URL from the response
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    with open(f"test{i+1}.jpg", 'wb') as handler:
        handler.write(img_data)
```

Import necessary modules: `os`, `openai`, `requests` are Python libraries used in the script, and secret is a local Python file containing sensitive data like API keys.

`openai.api_key=secret.api_key`: This line is used to set the API key required to authenticate requests to the OpenAI API. The API key is fetched from a secret Python file, which is not included in the script.

The chain function is defined to accept a list of prompts. For each prompt, an image is created using the `openai.Image.create()` method. This method makes a call to OpenAI's API using the provided prompt, and requests one image of size "256x256".

`image_url = response['data'][0]['url']`: Once the response is received from the API, the URL of the generated image is extracted from the response.

`img_data = requests.get(image_url).content`: This line is used to download the image data from the extracted URL using the requests library.

The script then opens a file in write-binary mode (wb) and writes the image data into the file. The image file is named "test1.jpg", "test2.jpg", etc., depending on the current iteration in the loop.

Check It!

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